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These notes are for ECON 4304: Time Series Econometrics and Business Forecasting at HKUST. The primary focus is on the mathematical derivation and analysis of time series models, and empirical applications are left for independent study. The content is adapted from Dr. Eric Ng's course materials, with the main differences being minor adjustments to the content sequence and a greater emphasis on matrix algebra.

1 Modeling Time Series Data: ARMA Models

1.1 Introduction

- What is a time series?
 - A natural ordering by dates: $1, 2, \dots, t, t+1, \dots, T$.
 - A time series is a sequence of observations ordered by dates: $\{y_1, y_2, \dots, y_t, y_{t+1}, \dots, y_T\}$.
- Motivation of time series analysis and modeling:
 - Understanding past data, forecasting future data.
- Some features in real-life time series:
 - Trend, Seasonality, Cyclicity, and Irregularity.
 - Current shock u_t tends to affect both present variable y_t and future variables $y_{t+1}, y_{t+2}, \dots$
 - Structural breaks (sudden changes over time) in mean, variance, and/or regression coefficients.
 - Spurious regression: Strong relationship between completely unrelated time series variables.
 - Autocorrelated: The present observation is correlated with both past and future observations:

$$\text{Cov}(y_{t-1}, y_t) \neq 0, \text{Cov}(y_t, y_{t+1}) \neq 0$$

1.2 Stationarity

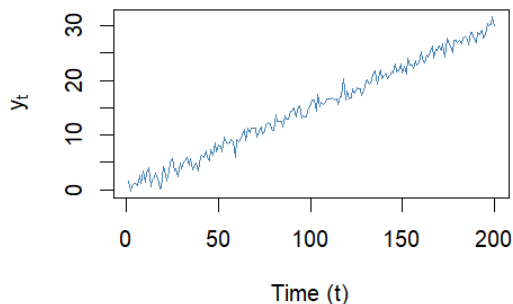
- Our models and forecasting primarily focus on **covariance stationary**, or simply **stationary** time series data, which has the following properties:

$$\mathbb{E}[y_t] = \mu < \infty$$

$$\mathbb{E}[(y_t - \mu)^2] = \text{Var}(y_t) < \infty$$

$$\mathbb{E}[(y_t - \mu)(y_{t+j} - \mu)] = \text{Cov}(y_t, y_{t+j}) = \gamma_j < \infty \text{ for all } j = 0, 1, 2, \dots$$

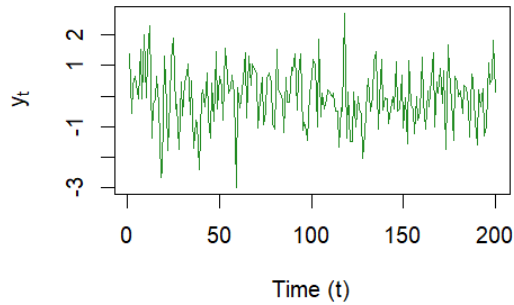
- γ_j is called **autocovariance**. Note that $\gamma_0 = \text{Var}(y_t)$.
- Mean, variance, and autocovariance do not change over time for a stationary process.
- For a stationary process, $\text{Cov}(y_{t-j}, y_t) = \text{Cov}(y_t, y_{t+j}) = \gamma_j$.
- Some examples of stationary/nonstationary processes:
 - A growing trend is not stationary as the mean is increasing over time.



- **White noise process:**

$$y_t = u_t$$

where $\mathbb{E}[u_t] = 0$, $\text{Var}(u_t) = \gamma_0 = \sigma^2$ and $\gamma_j = 0$ by definition, which is stationary. Oftentimes we take $u_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.

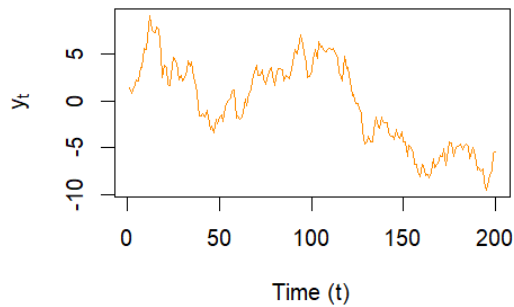


- **Random walk process (without drift):**

$$y_t = y_{t-1} + u_t$$

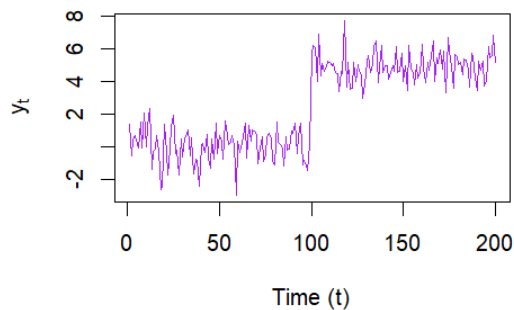
is not stationary, because:

$$\text{Var}(y_t) = \text{Var}(y_0 + \sum_{i=1}^t u_i) = \text{Var}(\sum_{i=1}^t u_i) = \sum_{i=1}^t \text{Var}(u_i) = t\sigma^2$$



- A time series with **structural break** is not stationary:

$$y_t = \begin{cases} \mu_1 + u_t & \text{if } t < T_0 \\ \mu_2 + u_t & \text{if } t \geq T_0 \end{cases}$$



1.3 AR model, ACF and PACF

- An **autoregressive model** with order p , $\text{AR}(p)$, is defined as:

$$y_t = c + \sum_{j=1}^p \phi_j y_{t-j} + u_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + u_t$$

where u_t is a white noise process.

- In practice, one of the most widely used model is $\text{AR}(1)$:

$$y_t = c + \phi y_{t-1} + u_t$$

- For an $\text{AR}(1)$ to be stationary, we must have $|\phi| < 1$, because:

$$\text{Var}(y_t) = \text{Var}(c + \phi y_{t-1} + u_t) = \phi^2 \text{Var}(y_{t-1}) + \sigma^2$$

For a stationary model, $\text{Var}(y_t) = \text{Var}(y_{t-1}) = \gamma_0$, thus:

$$\text{Var}(y_t) = \gamma_0 = \frac{\sigma^2}{1 - \phi^2}$$

Hence, $\text{Var}(y_t) > 0$ only if $|\phi| < 1$.

- For an $\text{AR}(p)$ to be stationary, all roots for the characteristic equation $z^p - \sum_{j=1}^p \phi_j z^{p-j} = 0$ must lie within a complex unit circle.
- Apart from the mean, variance and autocovariance of a time series, we are also naturally interested in the **autocorrelation**:

$$\rho_j = \frac{\text{Cov}(y_{t-j}, y_t)}{\text{Var}(y_t)}$$

- Computing ρ_j for all j , we can obtain the **autocorrelation function**, $\text{ACF}(j)$.
- For example, for an $\text{AR}(1)$: $y_t = c + \phi y_{t-1} + u_t$,

$$\begin{aligned} \gamma_1 &= \text{Cov}(y_t, y_{t-1}) = \text{Cov}(c + \phi y_{t-1} + u_t, y_{t-1}) = \phi \gamma_0 \\ \gamma_2 &= \text{Cov}(y_t, y_{t-2}) = \text{Cov}(c + \phi y_{t-1} + u_t, y_{t-2}) = \phi \gamma_1 = \phi^2 \gamma_0 \end{aligned}$$

Thus, we can deduce that:

$$\gamma_j = \phi^j \gamma_0$$

It follows that:

$$\rho_j = \frac{\gamma_j}{\gamma_0} = \phi^j$$

- For AR models with higher order, this method isn't as useful since different y_{t-j} terms are correlated.
- Instead, we can derive something called **Yule-Walker equations**. For simplicity, consider an $\text{AR}(2)$ with zero mean: $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t$, then we can make use of a property:

$$\gamma_j = \mathbb{E}[(y_t - \mu)(y_{t-j} - \mu)] = \mathbb{E}[y_t y_{t-j}]$$

We can derive that:

$$\begin{aligned} y_t y_{t-j} &= \phi_1 y_{t-1} y_{t-j} + \phi_2 y_{t-2} y_{t-j} + u_t y_{t-j} \\ \mathbb{E}[y_t y_{t-j}] &= \phi_1 \mathbb{E}[y_{t-1} y_{t-j}] + \phi_2 \mathbb{E}[y_{t-2} y_{t-j}] + \mathbb{E}[u_t y_{t-j}] \end{aligned}$$

Note that $\mathbb{E}[y_t y_{t-j}] = \mathbb{E}[y_{t+k} y_{t+k-j}] = \gamma_j$, $\mathbb{E}[u_t y_t] = \sigma^2$, and $\mathbb{E}[u_t y_{t-j}] = 0$. Hence:

$$\gamma_j = \mathbb{E}[y_t y_{t-j}] = \begin{cases} \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2 & \text{if } j = 0 \\ \phi_1 \gamma_0 + \phi_2 \gamma_1 & \text{if } j = 1 \\ \phi_1 \gamma_{j-1} + \phi_2 \gamma_{j-2} & \text{otherwise} \end{cases}$$

Dividing both sides by γ_0 , we can get $\rho_j = \frac{\gamma_j}{\gamma_0}$:

$$\begin{cases} \rho_0 = \phi_1 \rho_1 + \phi_2 \rho_2 + \frac{\sigma^2}{\gamma_0} = 1 \\ \rho_1 = \phi_1 + \phi_2 \rho_1 \\ \rho_j = \phi_1 \rho_{j-1} + \phi_2 \rho_{j-2} \text{ for all } j > 1 \end{cases}$$

Solving for $j = 1, 2$, we can get $\rho_1 = \frac{\phi_1}{1-\phi_2}$, $\rho_2 = \frac{\phi_1^2}{1-\phi_2} + \phi_2$, and the rest can be solved by iterated substitution in the last equation.

- The ACF for both AR(1) and AR(2) are persistent ($\neq 0$) for all j , why?
 - To find the association between y_{t-j} and y_t , there is always a path of propagation through $y_{t-j+1}, \dots, y_{t-1}$.
 - Is there a more “direct” way of measurement?
- We can study the **partial autocorrelation function**, PACF(j).
- For each lag j , we simply fit a AR(j) to y_t :

$$y_t = c + \phi_{j,1} y_{t-1} + \dots + \phi_{j,j} y_{t-j} + u_t$$

PACF(j) is defined as $\phi_{j,j}$, which can be understood as the direct association between y_{t-j} and y_t .

- For an AR(p), we look at several cases:

1. $j = 1$: The regression equation is:

$$y_t = c + \phi_{1,1} y_{t-1} + u_t$$

Then,

$$\text{Cov}(y_t, y_{t-1}) = \phi_{1,1} \text{Var}(y_{t-1}) \implies \phi_{1,1} = \frac{\text{Cov}(y_t, y_{t-1})}{\text{Var}(y_t)} = \rho_1$$

which implies PACF(1)=ACF(1) always.

2. $1 < j < p$: PACF(j) $\neq$ ACF(j) in general, because PACF takes away the effects of lower lags.
 3. $j = p$: The regression is equivalent to the true model AR(p), thus PACF(p) = $\phi_{p,p} = \phi_p$.
 4. $j > p$: Since the regression best fits to AR(p), the regression coefficients for all $j > p$ evaluate to 0, thus PACF(j) = 0 for all $j > p$.
- So far we have been discussing the **model-implied** ACF and PACF, what about the actual time series data, given by a dataset $\{y_1, \dots, y_T\}$?

- Like a regular dataset, we can compute some sample parameters, namely:

Sample mean:

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t$$

Sample autocovariance:

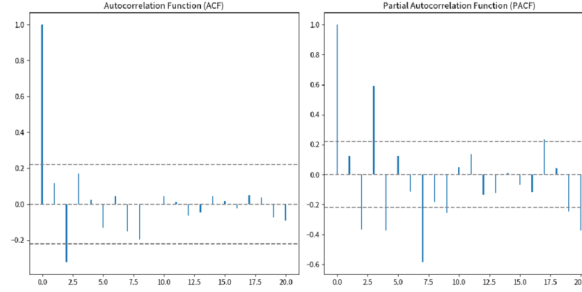
$$\hat{\gamma}_j = \frac{1}{T} \sum_{t=1}^{T-j} (y_{t+j} - \bar{y})(y_t - \bar{y})$$

Sample autocorrelation:

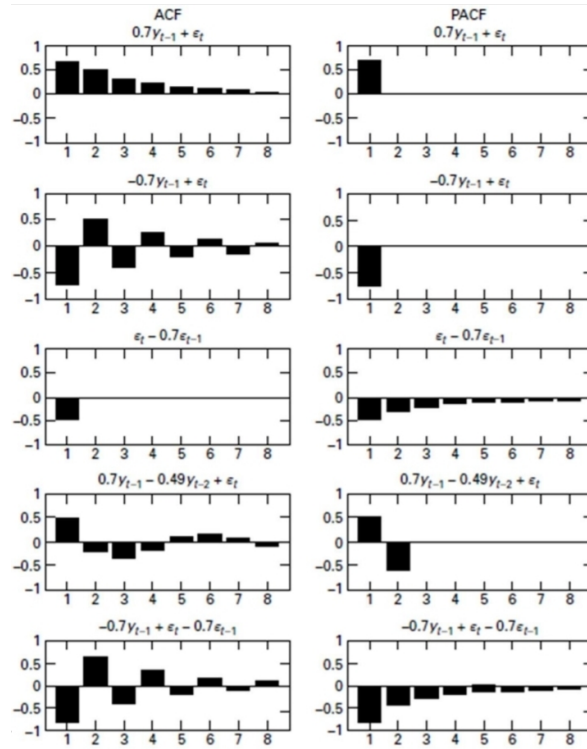
$$\hat{\rho}_j = \frac{\hat{\gamma}_j}{\hat{\gamma}_0}$$

which is the **sample ACF**.

- Sample PACF is still computed by regression.
- Once we have the sample ACF and PACF, we can fit our data to a model that has similar model-implied ACF and PACF. Specifically, we can look at the confidence interval to determine if they are significantly different from 0.



- Some examples of model-implied ACF and PACF, namely AR(1), AR(1), MA(1), AR(2), and ARMA(1,1):



- After fitting a model, we need to check whether it fulfills the assumption that the residuals $\hat{u}_t = y_t - \hat{y}_t$ are white noises.
- Both **Box-Pierce** and **Ljung-Box test** test for the following:
 - H_0 : The first m autocorrelations are zero ($\rho_1 = \dots = \rho_m = 0$).
 - H_1 : At least one autocorrelation is not zero.

- Then, we can look at the Q-statistic ($Q = T \sum_{j=1}^m \hat{\rho}_j^2 \sim \chi_{m-p}^2$ for Box-Pierce, $Q = T(T+2) \sum_{j=1}^m \frac{\hat{\rho}_j^2}{T-j} \sim \chi_{m-p}^2$ for Ljung-Box) and the corresponding p-value to determine whether H_0 is rejected. If H_0 is rejected, then the residuals are not white noises, which implies the model is not good enough.

1.4 Forecasting with AR Model

- We want to get the **j-step forecast** for y_{t+j} , given an **information set** $I_t = \{y_1, \dots, y_t, u_1, \dots, u_t\}$.
- Consider an AR(1): $y_t = c + \phi y_{t-1} + u_t$.
- Note that $\mathbb{E}[y_{t-j} | I_t] = y_{t-j}$ for all $j \geq 0$, and $\mathbb{E}[u_{t+j} | I_t] = 0$ for all $j > 0$.
- Then:

$$\begin{aligned}\mathbb{E}[y_{t+1} | I_t] &= \mathbb{E}[c + \phi y_t + u_{t+1} | I_t] = c + \phi y_t \\ \mathbb{E}[y_{t+2} | I_t] &= \mathbb{E}[c + \phi y_{t+1} + u_{t+2} | I_t] = c + \phi \mathbb{E}[y_{t+1} | I_t] = (1 + \phi)c + \phi^2 y_t\end{aligned}$$

It follows that:

$$\mathbb{E}[y_{t+j} | I_t] = c \sum_{k=0}^{j-1} \phi^k + \phi^j y_t$$

- The **long term forecast** is equivalent to the **unconditional forecast**:

$$\mathbb{E}[y_t] = \lim_{j \rightarrow \infty} \mathbb{E}[y_{t+j} | I_t] = c \sum_{k=0}^{\infty} \phi^k = \frac{c}{1 - \phi}$$

Alternatively, we can also directly solve algebraically, using the property $\mathbb{E}[y_t] = \mathbb{E}[y_{t-1}]$:

$$\mathbb{E}[y_t] = \mathbb{E}[c + \phi y_{t-1} + u_t] = c + \phi \mathbb{E}[y_t] \implies \mathbb{E}[y_t] = \frac{c}{1 - \phi}$$

- What is the forecast error?

$$e_t(1) = y_{t+1} - \mathbb{E}[y_{t+1} | I_t] = (c + \phi y_t + u_{t+1}) - (c + \phi y_t) = u_{t+1}$$

$$e_t(2) = y_{t+2} - \mathbb{E}[y_{t+2} | I_t] = (c + \phi y_{t+1} + u_{t+2}) - (c + \phi \mathbb{E}[y_{t+1} | I_t]) = u_{t+2} + \phi(y_{t+1} - \mathbb{E}[y_{t+1} | I_t]) = u_{t+2} + \phi u_{t+1}$$

It follows that:

$$e_t(j) = \sum_{k=1}^j \phi^{j-k} u_{t+k}$$

- What is the variance of the forecast error?

$$\text{Var}(e_t(j)) = \text{Var}\left(\sum_{k=1}^j \phi^{j-k} u_{t+k}\right) = \sum_{k=1}^j \phi^{2(j-k)} \text{Var}(u_t) = \sum_{k=1}^j \phi^{2(j-k)} \sigma^2$$

Let $m = j - k$, then $\sum_{k=1}^j \phi^{2(j-k)} \sigma^2 = \sum_{m=0}^{j-1} \phi^{2m} \sigma^2$. As $j \rightarrow \infty$:

$$\lim_{j \rightarrow \infty} \text{Var}(e_t(j)) = \sum_{m=0}^{\infty} \phi^{2m} \sigma^2 = \frac{\sigma^2}{1 - \phi^2} = \text{Var}(y_t)$$

- If u_t is i.i.d. normal, then $e_t(j)$ is also normal:

$$e_t(j) | I_t \sim \mathcal{N}(0, \text{Var}(e_t(j)))$$

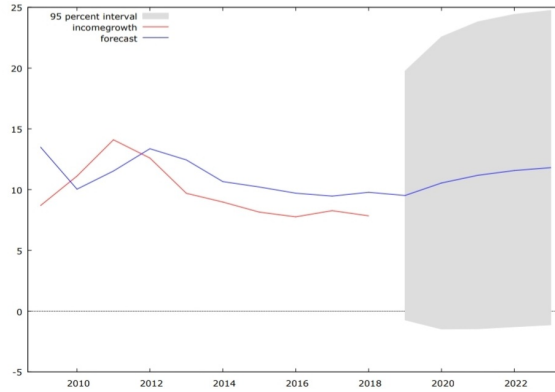
By standardization:

$$\mathbb{P} \left(\left| \frac{e_t(j)}{\sqrt{\text{Var}(e_t(j))}} \right| < 1.96 \mid I_t \right) = \mathbb{P} \left(\left| \frac{y_{t+j} - \mathbb{E}[y_{t+j} \mid I_t]}{\sqrt{\text{Var}(e_t(j))}} \right| < 1.96 \mid I_t \right) = 0.95$$

Hence, the 95% confidence interval for y_{t+j} is:

$$y_{t+j} \in \left[\mathbb{E}[y_{t+j} \mid I_t] \pm 1.96 \sqrt{\text{Var}(e_t(j))} \right]$$

- If the model is estimated using all data $\{y_1, \dots, y_T\}$, then there can be two kinds of forecasting:
 1. **In-sample forecasting:** $\mathbb{E}[y_{t+j} \mid I_t]$, where $t+j \leq T$. This serves as **backtesting** to see if the model aligns with the actual data in the past.
 2. **Out-of-sample forecasting:** $\mathbb{E}[y_{T+j} \mid I_T]$, where $j > 0$. The actual forecasting we want to make.
- In practice, for better evaluation, we can separate the data: Use $\{y_1, \dots, y_t\}$ as the **training data** to estimate the model, and use $\{y_{t+1}, \dots, y_T\}$ as the **test data** to evaluate the performance.



1.5 MA Model

- A **moving average** model with order q , $\text{MA}(q)$, is defined as:

$$y_t = \mu + u_t + \sum_{j=1}^q \theta_j u_{t-j} = \mu + u_t + \theta_1 u_{t-1} + \dots + \theta_q u_{t-q}$$

where u_t is a white noise process.

- Why is it called a moving average?
- An example:
 - Consider tossing a fair coin and let u_t denote the payoff at toss t , \$1 if wins and -\$1 if loses. Note that u_t fulfills all three conditions for a white noise. Let y_t be the average payoff on the last 4 tosses.

$$y_t = \frac{u_t + u_{t-1} + u_{t-2} + u_{t-3}}{4} = 0.25u_t + 0.25u_{t-1} + 0.25u_{t-2} + 0.25u_{t-3}$$

- An $\text{MA}(q)$ is always stationary, because:

$$\text{Var}(y_t) = \text{Var}(\mu + u_t + \theta_1 u_{t-1} + \dots + \theta_q u_{t-q}) = \text{Var}(u_t) + \theta_1^2 \text{Var}(u_{t-1}) + \dots + \theta_q^2 \text{Var}(u_{t-q}) = (1 + \sum_{j=1}^q \theta_j^2) \sigma^2$$

- Consider an MA(1) with zero mean: $y_t = u_t + \theta u_{t-1}$. Again, we have $\gamma_j = \mathbb{E}[y_t y_{t-j}]$. We derive the ACF by the Yule-Walker equations.

$$\begin{aligned} y_t y_{t-j} &= u_t y_{t-j} + \theta u_{t-1} y_{t-j} \\ \mathbb{E}[y_t y_{t-j}] &= \mathbb{E}[u_t y_{t-j}] + \theta \mathbb{E}[u_{t-1} y_{t-j}] \\ &= \mathbb{E}[u_t (u_{t-j} + \theta u_{t-j-1})] + \theta \mathbb{E}[u_{t-1} (u_{t-j} + \theta u_{t-j-1})] \\ &= \mathbb{E}[u_t u_{t-j}] + \theta \mathbb{E}[u_t u_{t-j-1}] + \theta \mathbb{E}[u_{t-1} u_{t-j}] + \theta^2 \mathbb{E}[u_{t-1} u_{t-j-1}] \end{aligned}$$

Note that $\mathbb{E}[u_t^2] = \sigma^2$ and $\mathbb{E}[u_t u_{t-j}] = 0$ for all $j \neq 0$ by definition of white noise. Then, for $j = 0$:

$$\gamma_0 = \mathbb{E}[y_t^2] = (1 + \theta^2)\sigma^2$$

For $j = 1$:

$$\gamma_1 = \mathbb{E}[y_t y_{t-1}] = \theta \sigma^2$$

And $\gamma_j = 0$ for all $j \geq 2$. Dividing both sides by γ_0 , the ACF is simply:

$$\rho_j = \frac{\gamma_j}{\gamma_0} = \begin{cases} \frac{\theta}{1+\theta^2} & \text{if } j = 1 \\ 0 & \text{otherwise} \end{cases}$$

- The PACF for an MA(q) is a long tail (persists infinitely), since it is equivalent to an AR(∞) (under certain conditions, proof omitted).
- Conversely, an AR(p) is equivalent to an MA(∞). For example, for an AR(1) :

$$y_t = c + \phi y_{t-1} + u_t = c + \phi(c + \phi y_{t-2} + u_{t-1}) + u_t = \dots = \frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j u_{t-j}$$

- This is the reason why their ACF and PACF have opposite shapes.

1.6 ARMA Model

- An **autoregressive moving average model** with order (p, q) , ARMA(p, q), is defined as:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + u_t + \sum_{j=1}^q \theta_j u_{t-j}$$

- Stationarity only depends on the AR coefficients with the same condition: $|\phi| < 1$ for $p = 1$, all roots for the characteristic equation lie within a unit circle for $p \geq 1$.
- Wold's decomposition theorem: Any stationary series has an MA(∞) representation, including ARMA(p, q), AR(p), MA(q).
- Yule-Walker equations for an ARMA(1,1) with zero mean: $y_t = \phi y_{t-1} + u_t + \theta u_{t-1}$,

$$\begin{aligned} y_t y_{t-j} &= \phi y_{t-1} y_{t-j} + u_t y_{t-j} + \theta u_{t-1} y_{t-j} \\ \mathbb{E}[y_t y_{t-j}] &= \phi \mathbb{E}[y_{t-1} y_{t-j}] + \mathbb{E}[u_t y_{t-j}] + \theta \mathbb{E}[u_{t-1} y_{t-j}] \end{aligned}$$

Note that $\mathbb{E}[y_{t-1} y_{t-j}] = \gamma_{j-1}$, $\mathbb{E}[u_t y_t] = \sigma^2$, and $\mathbb{E}[u_t y_{t-j}] = 0$ for $j > 0$. Also,

$$\mathbb{E}[u_{t-1} y_t] = \mathbb{E}[u_{t-1} (\phi y_{t-1} + u_t + \theta u_{t-1})] = (\phi + \theta)\sigma^2$$

Then, we have:

$$\begin{cases} \gamma_0 = \phi \gamma_1 + \sigma^2 + \theta(\phi + \theta)\sigma^2 \\ \gamma_1 = \phi \gamma_0 + \theta \sigma^2 \\ \gamma_j = \phi \gamma_{j-1} \end{cases} \quad \text{for all } j \geq 2$$

Solving for the first two equations, we can get $\gamma_0 = \frac{1+\theta^2+2\phi\theta}{1-\phi^2}\sigma^2$ and $\gamma_1 = \frac{(1+\phi\theta)(\phi+\theta)}{1-\phi^2}\sigma^2$. Hence, $\rho_1 = \frac{(1+\phi\theta)(\phi+\theta)}{1+\theta^2+2\phi\theta}$ and $\rho_j = \phi \rho_{j-1}$ for all $j \geq 2$.

1.7 Model Selection

- While we can choose the parameters (p, q) by looking at the sample ACF and PACF, often times we still have multiple candidates.
- We can use something called **information criteria**, for example:

$$\begin{aligned} \text{AIC} &= 2 \ln(\hat{\delta}) + \frac{2k}{T} \\ \text{SBIC} &= 2 \ln(\hat{\delta}) + \frac{k \ln(T)}{T} \\ \text{HBIC} &= 2 \ln(\hat{\delta}) + \frac{2k \ln(\ln(T))}{T} \end{aligned}$$

where $\hat{\delta}$ is the error (e.g. SER, RMSE) and k is the number of parameters (i.e. c, ϕ_i, θ_j).

- Other things being equal, we should choose a model with the smallest information criteria, which gives us a model with the best trade-off between bias (underfitting) and variance (overfitting).

2 Modeling Conditional Variance: ARCH and GARCH Models

2.1 Model Specifications

- So far we have mainly been using the **unconditional variance** of the error term: $\text{Var}(u_t) = \sigma^2$, but observe that when we compute the variance of a conditional forecast, say, for an ARMA(1,1):

$$\text{Var}(y_{t+1} | I_t) = \text{Var}(c + \phi y_t + u_{t+1} + \theta u_t | I_t) = \text{Var}(u_{t+1} | I_t)$$

we have to make use of the **conditional variance** $\text{Var}(u_{t+1} | I_t)$.

- It is natural to think that it is constant: $\text{Var}(u_{t+1} | I_t) = \text{Var}(u_t) = \sigma^2$ (**homoscedastic**), but is it really the case?
- For example, stock returns are often **heteroscedastic** and exhibit the following features:
 1. Volatility cluster: If the volatility in period $t-1$ is high, then it's more likely that period t is also volatile.
 2. Fat tails: The probability of having extremely high or extremely low returns (kurtosis) is higher than usual.
 3. Asymmetry: Negative shocks at period $t-1$ have a stronger effect on period t .
- In addition to the white noise assumptions for u_t , we strengthen it with one more condition:

$$\mathbb{E}[u_t | I_{t-1}] = 0$$

which makes it a **martingale difference sequence**.

- Then, we can obtain:

$$\text{Var}(y_t | I_{t-1}) = \text{Var}(u_t | I_{t-1}) = \mathbb{E}[u_t^2 | I_{t-1}] - \mathbb{E}[u_t | I_{t-1}]^2 = \mathbb{E}[u_t^2 | I_{t-1}]$$

To model the volatility cluster, we may assume:

$$\mathbb{E}[u_t^2 | I_{t-1}] = \alpha_0 + \sum_{j=1}^q \alpha_j u_{t-j}^2$$

- One idea is using an AR model:

$$u_t^2 = \alpha_0 + \sum_{j=1}^q \alpha_j u_{t-j}^2 + v_t$$

where $\mathbb{E}[v_t | I_{t-1}] = 0$.

- Engle (1982) proposed the **autoregressive conditional heteroskedasticity** model, ARCH(q), instead, defined as:

$$u_t = v_t \sqrt{h_t}, \quad h_t = \alpha_0 + \sum_{j=1}^q \alpha_j u_{t-j}^2$$

where $\mathbb{E}[v_t | I_{t-1}] = 0$ and $\mathbb{E}[v_t^2 | I_{t-1}] = 1$. For example, we can assume $v_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$.

- Note that u_t and h_t are uncorrelated. Then, we can derive some features:

- Volatility cluster: (Note that since $u_{t-j} \in I_{t-1}$ for all $j \geq 1$, $h_t \in I_{t-1}$.)

$$\mathbb{E}[u_t^2 | I_{t-1}] = \mathbb{E}[v_t^2 h_t | I_{t-1}] = \mathbb{E}[v_t^2 | I_{t-1}] \cdot h_t = h_t$$

Since we are interested in the conditional variance, $\text{Var}(y_t | I_{t-1}) = \mathbb{E}[u_t^2 | I_{t-1}] = h_t$ means we are essentially modeling h_t .

- Martingale difference:

$$\mathbb{E}[u_t | I_{t-1}] = \mathbb{E}[v_t \sqrt{h_t} | I_{t-1}] = \mathbb{E}[v_t | I_{t-1}] \mathbb{E}[\sqrt{h_t} | I_{t-1}] = 0$$

- No serial correlation:

$$\mathbb{E}[u_t u_{t-j} | I_{t-1}] = \mathbb{E}[u_t | I_{t-1}] \cdot u_{t-j} = 0$$

- Stationary (Mean reversion): (Note that the unconditional variance $\mathbb{E}[u_t^2] = \mathbb{E}[u_{t-j}^2]$.)

$$\mathbb{E}[u_t^2] = \mathbb{E}[\mathbb{E}[u_t^2 | I_{t-1}]] = \mathbb{E}[\alpha_0 + \sum_{j=1}^q \alpha_j u_{t-j}^2] = \alpha_0 + \sum_{j=1}^q \alpha_j \mathbb{E}[u_t^2] \implies \mathbb{E}[u_t^2] = \frac{\alpha_0}{1 - \sum_{j=1}^q \alpha_j}$$

Therefore, u_t is stationary only if $\sum_{j=1}^q \alpha_j < 1$.

- We can also verify that u_t is still a white noise process, which is compatible with an ARMA(p, q).

- We also have the generalized ARCH, GARCH(p, q), defined as:

$$u_t = v_t \sqrt{h_t}, \quad h_t = \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}$$

where $\mathbb{E}[v_t | I_{t-1}] = 0$ and $\mathbb{E}[v_t^2 | I_{t-1}] = 1$.

- In a GARCH model, conditional variance depends not only on past shocks, but also on past variances. Thus, it captures persistent volatility more effectively. For example, modeling highly persistent volatility often requires an ARCH(q) with many lags, whereas a GARCH(1,1) suffices.

- GARCH also has similar features, for example:

- Conditional volatility:

$$\mathbb{E}[u_t^2 | I_{t-1}] = \mathbb{E}[v_t^2 h_t | I_{t-1}] = \mathbb{E}[v_t^2 | I_{t-1}] \cdot h_t = h_t$$

- For unconditional volatility, first note that:

$$\mathbb{E}[u_t^2] = \mathbb{E}[\mathbb{E}[u_t^2 | I_{t-1}]] = \mathbb{E}[h_t]$$

Also given that $\mathbb{E}[u_t^2] = \mathbb{E}[u_{t-j}^2]$ and $\mathbb{E}[h_t] = \mathbb{E}[h_{t-j}]$, we can get:

$$\mathbb{E} \left[\alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j} \right] = \alpha_0 + \sum_{i=1}^q \alpha_i \mathbb{E}[u_t^2] + \sum_{j=1}^p \beta_j \mathbb{E}[h_t] = \alpha_0 + \sum_{i=1}^q \alpha_i \mathbb{E}[u_t^2] + \sum_{j=1}^p \beta_j \mathbb{E}[u_t^2]$$

Therefore:

$$\mathbb{E}[u_t^2] = \frac{\alpha_0}{1 - \sum_{i=1}^q \alpha_i - \sum_{j=1}^p \beta_j}$$

- Hence, u_t is stationary only if $\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j < 1$.

- To model asymmetry, we may use a GJR-GARCH model, defined as:

$$u_t = v_t \sqrt{h_t}, \quad h_t = \alpha_0 + \sum_{i=1}^q (\alpha_i + \gamma_i \mathbb{I}(u_{t-i} < 0)) u_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}$$

where $\mathbb{I}(u_{t-i} < 0) = 1$ if $u_{t-i} < 0$ and 0 otherwise.

2.2 Forecasting with ARCH and GARCH

- Consider an ARCH(1):

$$u_t = v_t \sqrt{\alpha_0 + \alpha_1 u_{t-1}^2}$$

We will make use of the generalized law of iterated expectation:

$$\mathbb{E}[x_t | I_{t-j}] = \mathbb{E}[\mathbb{E}[x_t | I_{t-i}] | I_{t-j}]$$

if $j \geq i$. Since I_{t-j} is a smaller information set ($I_{t-j} \subset I_{t-i}$), it dominates the expectation.

First:

$$\mathbb{E}[u_{t+1}^2 | I_t] = \mathbb{E}[v_{t+1}^2 (\alpha_0 + \alpha_1 u_t^2) | I_t] = \mathbb{E}[v_{t+1}^2 | I_t] \cdot (\alpha_0 + \alpha_1 u_t^2) = \alpha_0 + \alpha_1 u_t^2$$

Thus, we know that:

$$\mathbb{E}[u_{t+2}^2 | I_{t+1}] = \alpha_0 + \alpha_1 u_{t+1}^2$$

Then:

$$\mathbb{E}[u_{t+2}^2 | I_t] = \mathbb{E}[\mathbb{E}[u_{t+2}^2 | I_{t+1}] | I_t] = \mathbb{E}[\alpha_0 + \alpha_1 u_{t+1}^2 | I_t] = \alpha_0 + \alpha_1 \mathbb{E}[u_{t+1}^2 | I_t]$$

This gives us a recurrence relation:

$$\mathbb{E}[u_{t+j}^2 | I_t] = \alpha_0 + \alpha_1 \mathbb{E}[u_{t+j-1}^2 | I_t]$$

which can be solved by iterated substitution till we reach $j = 1$.

- Similarly, for a GARCH(1,1):

$$u_t = v_t \sqrt{h_t}, \quad h_t = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta h_{t-1}$$

Again, note that $h_{t+1} \in I_t$. First:

$$\mathbb{E}[u_{t+1}^2 | I_t] = \mathbb{E}[v_{t+1}^2 h_{t+1} | I_t] = \mathbb{E}[v_{t+1}^2 | I_t] \cdot h_{t+1} = h_{t+1}$$

Thus, we know that:

$$\mathbb{E}[u_{t+2}^2 | I_{t+1}] = h_{t+2}$$

Then:

$$\mathbb{E}[u_{t+2}^2 | I_t] = \mathbb{E}[\mathbb{E}[u_{t+2}^2 | I_{t+1}] | I_t] = \mathbb{E}[h_{t+2} | I_t] = \mathbb{E}[\alpha_0 + \alpha_1 u_{t+1}^2 + \beta h_{t+1} | I_t]$$

Given that $\mathbb{E}[u_{t+1}^2 | I_t] = h_{t+1} = \mathbb{E}[h_{t+1} | I_t]$, we have:

$$\mathbb{E}[\alpha_0 + \alpha_1 u_{t+1}^2 + \beta h_{t+1} | I_t] = \alpha_0 + \alpha_1 \mathbb{E}[u_{t+1}^2 | I_t] + \beta \mathbb{E}[u_{t+1}^2 | I_t] = \alpha_0 + (\alpha_1 + \beta) \mathbb{E}[u_{t+1}^2 | I_t]$$

It follows that:

$$\mathbb{E}[u_{t+j}^2 | I_t] = \alpha_0 + (\alpha_1 + \beta) \mathbb{E}[u_{t+j-1}^2 | I_t]$$

2.3 Model Fitting and Diagnostics

- In practice, the error terms we want to model are derived from the residuals of another model. For example, a full model can be an AR(1)-GARCH(1,1):

$$\begin{aligned}y_t &= c + \phi y_{t-1} + u_t \\u_t &= v_t \sqrt{h_t} \\h_t &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta h_{t-1}\end{aligned}$$

where $\mathbb{E}[v_t | I_{t-1}] = 0$ and $\mathbb{E}[v_t^2 | I_{t-1}] = 1$.

- To choose the hyperparameters (p, q) for ARCH or GARCH, one can look at the ACF and PACF of $\hat{u}_t^2$, which is an estimator for h_t . Information criteria are also useful.
- After fitting and estimating the parameters $c, \phi, \alpha_0, \alpha_1, \beta$, we need to conduct autocorrelation test (Box–Pierce/Ljung–Box) for two items to ensure they have no autocorrelation:
 - $\hat{v}_t = \frac{\hat{u}_t}{\sqrt{\hat{h}_t}}$: If v_t is not a white noise, then u_t is not a white noise, which violates the AR assumption.
 - $\hat{v}_t^2 = \frac{\hat{u}_t^2}{\hat{h}_t}$: If $\hat{v}_t^2$ has serial autocorrelations, then there is remaining conditional heteroskedasticity that the GARCH cannot capture.

3 Modeling Multivariate Data: VAR Model

3.1 Structural VAR

- In ARMA models, we only focus on one variable y_t . However, what if we have multiple time series variables that are dependent on each other?
- For example, suppose y_{1t} follows AR(1), so it is dependent on $y_{1,t-1}$. On top of that, it is also dependent on another variable, y_{2t} and $y_{2,t-1}$, where y_{2t} follows a symmetrical relationship.

$$\begin{aligned}y_{1t} &= c_1 + \beta y_{2t} + \phi_{11} y_{1,t-1} + \phi_{12} y_{2,t-1} + \varepsilon_{1t} \\y_{2t} &= c_2 + \gamma y_{1t} + \phi_{21} y_{1,t-1} + \phi_{22} y_{2,t-1} + \varepsilon_{2t}\end{aligned}$$

- ε_{1t} and ε_{2t} are white noises and independent **structural shocks**: $\text{Cov}(\varepsilon_{1t}, \varepsilon_{2t}) = 0$.
- However, if $\beta, \gamma \neq 0$, then $\text{Cov}(y_{1t}, \varepsilon_{2t}), \text{Cov}(y_{2t}, \varepsilon_{1t}) \neq 0$ by a chain of reaction: $\varepsilon_{1t} \rightarrow y_{1t} \rightarrow y_{2t} \rightarrow y_{1t} \rightarrow y_{2t} \dots$
- Specifically:

$$\begin{aligned}\text{Cov}(y_{1t}, \varepsilon_{2t}) &= \text{Cov}(c_1 + \beta y_{2t} + \phi_{11} y_{1,t-1} + \phi_{12} y_{2,t-1} + \varepsilon_{1t}, \varepsilon_{2t}) \\&= \beta \text{Cov}(y_{2t}, \varepsilon_{2t}) \\&= \beta \text{Cov}(c_2 + \gamma y_{1t} + \phi_{21} y_{1,t-1} + \phi_{22} y_{2,t-1} + \varepsilon_{2t}, \varepsilon_{2t}) \\&= \beta \gamma \text{Cov}(y_{1t}, \varepsilon_{2t}) + \beta \text{Var}(\varepsilon_{2t}) \\&\implies \text{Cov}(y_{1t}, \varepsilon_{2t}) = \frac{\beta \text{Var}(\varepsilon_{2t})}{1 - \beta \gamma}\end{aligned}$$

By symmetry, we can get:

$$\text{Cov}(y_{2t}, \varepsilon_{1t}) = \frac{\gamma \text{Var}(\varepsilon_{1t})}{1 - \beta \gamma}$$

- Since the regressor is correlated with the error term, OLS fails to estimate this model.
- We call this form of model a **structural VAR** (vector autoregression) or **SVAR**. This can also be represented in a vector-matrix form:

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} 0 & \beta \\ \gamma & 0 \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

3.2 Reduced-form VAR

- For the model to be able to be estimated by OLS, we need to get rid of the y_{jt} term in the equation of y_{it} to obtain a **reduced-form VAR**.
- For example:

$$\begin{aligned}
 y_{1t} &= c_1 + \beta y_{2t} + \phi_{11}y_{1,t-1} + \phi_{12}y_{2,t-1} + \varepsilon_{1t} \\
 &= c_1 + \beta(c_2 + \gamma y_{1t} + \phi_{21}y_{1,t-1} + \phi_{22}y_{2,t-1} + \varepsilon_{2t}) + \phi_{11}y_{1,t-1} + \phi_{12}y_{2,t-1} + \varepsilon_{1t} \\
 &= (c_1 + \beta c_2) + \beta\gamma y_{1t} + (\phi_{11} + \beta\phi_{21})y_{1,t-1} + (\phi_{12} + \beta\phi_{22})y_{2,t-1} + (\varepsilon_{1t} + \beta\varepsilon_{2t}) \\
 \Rightarrow y_{1t} &= \frac{c_1 + \beta c_2}{1 - \beta\gamma} + \frac{\phi_{11} + \beta\phi_{21}}{1 - \beta\gamma}y_{1,t-1} + \frac{\phi_{12} + \beta\phi_{22}}{1 - \beta\gamma}y_{2,t-1} + \frac{\varepsilon_{1t} + \beta\varepsilon_{2t}}{1 - \beta\gamma}
 \end{aligned}$$

By symmetry:

$$y_{2t} = \frac{c_2 + \gamma c_1}{1 - \beta\gamma} + \frac{\phi_{21} + \gamma\phi_{11}}{1 - \beta\gamma}y_{1,t-1} + \frac{\phi_{22} + \gamma\phi_{12}}{1 - \beta\gamma}y_{2,t-1} + \frac{\varepsilon_{2t} + \gamma\varepsilon_{1t}}{1 - \beta\gamma}$$

Let $e_{1t} = \frac{\varepsilon_{1t} + \beta\varepsilon_{2t}}{1 - \beta\gamma}$ and $e_{2t} = \frac{\varepsilon_{2t} + \gamma\varepsilon_{1t}}{1 - \beta\gamma}$, since they are uncorrelated to any regressor ($y_{1,t-1}$, $y_{2,t-1}$), these two can be estimated by OLS. However, while $\text{Cov}(\varepsilon_{1t}, \varepsilon_{2t}) = 0$, now $\text{Cov}(e_{1t}, e_{2t}) \neq 0$.

- In vector-matrix form:

$$\begin{aligned}
 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} &= \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} 0 & \beta \\ \gamma & 0 \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix} \\
 \begin{bmatrix} 1 & -\beta \\ -\gamma & 1 \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} &= \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}
 \end{aligned}$$

We can denote this as:

$$\begin{aligned}
 \mathbf{A}_0 \mathbf{y}_t &= \mathbf{c} + \mathbf{\Psi} \mathbf{y}_{t-1} + \mathbf{\varepsilon}_t \\
 \mathbf{y}_t &= \mathbf{A}_0^{-1} \mathbf{c} + \mathbf{A}_0^{-1} \mathbf{\Psi} \mathbf{y}_{t-1} + \mathbf{A}_0^{-1} \mathbf{\varepsilon}_t
 \end{aligned}$$

Let $\mathbf{A} = \mathbf{A}_0^{-1}$, then we have $\mathbf{e}_t = \mathbf{A} \mathbf{\varepsilon}_t$.

- Recall the variance-covariance matrix of $\mathbf{x} = [x_1, x_2]^T$ is defined as:

$$\mathbf{\Sigma}_{\mathbf{x}} = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix}$$

If $\mathbf{y} = \mathbf{A} \mathbf{x}$, then $\mathbf{\Sigma}_{\mathbf{y}} = \mathbf{A} \mathbf{\Sigma}_{\mathbf{x}} \mathbf{A}^T$. Applying this to $\mathbf{e}_t$, we have $\mathbf{\Sigma}_{\mathbf{e}_t} = \mathbf{A} \mathbf{\Sigma}_{\mathbf{\varepsilon}_t} \mathbf{A}^T$. Since it is not a diagonal matrix, we know $\text{Cov}(e_{1t}, e_{2t}) \neq 0$.

- Comparing SVAR and reduced-form VAR:
 - An SVAR carries more economic meanings: variables can have contemporaneous effects on each other, error terms (structural shocks) are independent of each other. However, it cannot be estimated by OLS unless we add some constraints (discuss later).
 - A reduced-form VAR is more mathematically convenient and flexible to use.
- AR and VAR models can be generalized in a **companion form** which looks like a reduced-form VAR(1), expressed as:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{\Phi} \mathbf{y}_{t-1} + \mathbf{e}_t$$

where $\mathbf{\Phi}$ is called the **companion matrix**. For example:

- For an AR(p):

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + u_t$$

We may also consider the identities for all $1 < j < p$:

$$y_{t-j} = 0 + 1 \cdot y_{t-j} + 0$$

Then:

$$\begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix} = \begin{bmatrix} c \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} \phi_1 & \phi_2 & \dots & \phi_{p-1} & \phi_p \\ 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p} \end{bmatrix} + \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

- Similarly, for a reduced-form VAR(p):

$$\mathbf{y}_t = \mathbf{c} + \Phi_1 \mathbf{y}_{t-1} + \Phi_2 \mathbf{y}_{t-2} + \dots + \Phi_p \mathbf{y}_{t-p} + \mathbf{e}_t$$

where $\mathbf{y}_t = [y_{1t}, \dots, y_{nt}]^T$. We can simply stack the vectors and matrices together:

$$\begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix} = \begin{bmatrix} \mathbf{c} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \Phi_1 & \Phi_2 & \dots & \Phi_{p-1} & \Phi_p \\ \mathbf{I} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p} \end{bmatrix} + \begin{bmatrix} \mathbf{e}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}$$

- This form is helpful for checking stationarity: a VAR is stationary if and only if all eigenvalues of the companion matrix lie within a unit circle. One can verify that this aligns with the stationarity condition for AR(p) defined by the characteristic equation previously.

3.3 Recursive VAR

- An SVAR cannot be estimated by OLS but a reduced-form VAR can. Then, is it possible that we first estimate a reduced-form VAR, and then transform it back to an SVAR?
- First, we use a companion form to represent any reduced-form VAR:

$$\mathbf{y}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \mathbf{e}_t$$

We assume that the error term $\mathbf{e}_t$ is derived from an unknown SVAR, where $\mathbf{e}_t = \mathbf{A} \boldsymbol{\varepsilon}_t$. Furthermore, for easier calculation later on, we can consider a **normalized** error term $\tilde{\boldsymbol{\varepsilon}}_t = \boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}_t}^{-\frac{1}{2}} \boldsymbol{\varepsilon}_t$ such that $\boldsymbol{\Sigma}_{\tilde{\boldsymbol{\varepsilon}}_t} = \boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}_t}^{-\frac{1}{2}} \boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}_t} (\boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}_t}^{-\frac{1}{2}})^T = \mathbf{I}$. Then, we can rewrite the reduced-form VAR back into an SVAR:

$$\mathbf{y}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \mathbf{e}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \mathbf{A} \boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}_t}^{\frac{1}{2}} \tilde{\boldsymbol{\varepsilon}}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \tilde{\mathbf{A}} \tilde{\boldsymbol{\varepsilon}}_t$$

To transform back into an SVAR, we now want to find $\tilde{\mathbf{A}}$ from the reduced-form VAR.

- Also note that since all $\tilde{\varepsilon}_{j,t}$ have a standard deviation of one, a one-unit increase in $\tilde{\varepsilon}_{j,t}$ essentially means a one-SD shock in $y_{j,t}$, which is often more useful than the raw $\varepsilon_{j,t}$ in analysis.
- For example, suppose we first estimated a reduced-form VAR and we want to turn it back to an SVAR:

$$\mathbf{y}_t = \Phi \mathbf{y}_{t-1} + \mathbf{e}_t = \Phi \mathbf{y}_{t-1} + \tilde{\mathbf{A}} \tilde{\boldsymbol{\varepsilon}}_t$$

To get $\tilde{\mathbf{A}}$, we rely on the variance-covariance matrix of $\mathbf{e}_t$, where:

$$\boldsymbol{\Sigma}_{\mathbf{e}_t} = \tilde{\mathbf{A}} \boldsymbol{\Sigma}_{\tilde{\boldsymbol{\varepsilon}}_t} \tilde{\mathbf{A}}^T = \tilde{\mathbf{A}} \tilde{\mathbf{A}}^T$$

Expanding for a two-variable case, we have:

$$\begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

which gives us the following equations:

$$\begin{cases} \sigma_1^2 = a_{11}^2 + a_{12}^2 \\ \sigma_{12} = a_{11}a_{21} + a_{12}a_{22} \\ \sigma_{21} = a_{11}a_{21} + a_{12}a_{22} \\ \sigma_2^2 = a_{21}^2 + a_{22}^2 \end{cases}$$

Because $\sigma_{12} = \sigma_{21}$, the 2nd and 3rd equations are the same, which means we have 4 variables to be identified but only 3 equations. This is a **under-identification problem**.

- A simple solution: Let $a_{12} = 0$. This way, we assume y_1 is not affected by $\tilde{\varepsilon}_{2t}$, i.e. no contemporaneous effect.
- Generalizing to more variables, we assume $\tilde{\mathbf{A}}$ to be a lower triangular matrix to reduce the number of variables.
 - We need to rank the variables from the most exogenous to the least, that is, y_{1t} is only affected by $\tilde{\varepsilon}_{1t}$ contemporaneously, y_{2t} is only affected by $\tilde{\varepsilon}_{1t}$ and $\tilde{\varepsilon}_{2t}$, $\dots$, and $y_{n,t}$ is affected by all structural shocks.
 - This form is called a **recursive VAR**.
 - The order is usually implied by theory or observations. For example, consider monetary policy: We may assume inflation π_t is given exogenously, unemployment u_t is affected by it by the Phillips curve, and finally interest rate i_t is determined by both of these through Taylor rule.
- Then, we can solve for:

$$\begin{bmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_n^2 \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} & a_{31} & \dots & a_{n1} \\ 0 & a_{22} & a_{32} & \dots & a_{n2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{nn} \end{bmatrix}$$

To do this, we can apply a method called **Cholesky decomposition**, which can decompose a symmetrical matrix $\mathbf{A}$ as $\mathbf{A} = \mathbf{L}\mathbf{L}^T$, where $\mathbf{L}$ is a lower triangular matrix. Applying this, we can obtain $\Sigma_{\mathbf{e}_t} = \tilde{\mathbf{A}}\tilde{\mathbf{A}}^T$.

3.4 Impulse Responses

- Sometimes we don't just concern forecasting from the current state, but also what will happen if there is a structural shock.
 - For example, we don't just look at how the inflation and unemployment rates will go under the current interest rate, but rather how they will go if the Fed raises the interest rate by one unit today.
- We use a **impulse response function (IRF)** to represent this, defined as:

$$\text{IRF}(s) = \frac{\partial y_{i,t+s}}{\partial \varepsilon_{j,t}} = \frac{\partial y_{i,t}}{\partial \varepsilon_{j,t-s}}$$

- For example, for an AR(1), recall that it has an MA(∞) representation:

$$y_t = c + \phi y_{t-1} + \varepsilon_t = \frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$$

Thus:

$$\text{IRF}(s) = \frac{\partial y_t}{\partial \varepsilon_{t-s}} = \phi^s$$

- In a VAR model, while a shock in $\varepsilon_{j,t}$ only affects $y_{j,t}$ in period t , it propagates to all $y_{i,t+s}$ in the future.
- Similar to an AR(1), a VAR(1) also has a VMA(∞) representation:

$$\mathbf{y}_t = \mathbf{c} + \Phi \mathbf{y}_{t-1} + \mathbf{e}_t = \mathbf{c} + \Phi(\mathbf{c} + \Phi \mathbf{y}_{t-2} + \mathbf{e}_{t-1}) + \mathbf{e}_t = \dots = \mathbf{k} + \sum_{j=0}^{\infty} \Phi^j \mathbf{e}_{t-j} = \mathbf{k} + \sum_{j=0}^{\infty} \Phi^j \tilde{\mathbf{A}} \tilde{\mathbf{e}}_{t-j}$$

where $\mathbf{k} = \sum_{j=0}^{\infty} \Phi^j \mathbf{c}$ is a constant vector. Let $\mathbf{M}_j = \Phi^j \tilde{\mathbf{A}}$, then:

$$\mathbf{y}_t = \mathbf{k} + \sum_{j=0}^{\infty} \mathbf{M}_j \tilde{\mathbf{e}}_{t-j} = \mathbf{k} + \mathbf{M}_0 \tilde{\mathbf{e}}_t + \mathbf{M}_1 \tilde{\mathbf{e}}_{t-1} + \dots$$

We can see that a change in any $\varepsilon_{i,t-s}$ will only affect $\mathbf{y}_t$ through $\mathbf{M}_s$. Specifically, the IRF for $\mathbf{y}_{i,t+s}$ with respect to $\tilde{\varepsilon}_{j,t}$ is the (i,j) -element of $\mathbf{M}_s$. For example:

$$\begin{aligned} \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \end{bmatrix} &= \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} + \begin{bmatrix} m_{0,11} & m_{0,12} & m_{0,13} \\ m_{0,21} & m_{0,22} & m_{0,23} \\ m_{0,31} & m_{0,32} & m_{0,33} \end{bmatrix} \begin{bmatrix} \tilde{\varepsilon}_{1t} \\ \tilde{\varepsilon}_{2t} \\ \tilde{\varepsilon}_{3t} \end{bmatrix} + \begin{bmatrix} m_{1,11} & m_{1,12} & m_{1,13} \\ m_{1,21} & m_{1,22} & m_{1,23} \\ m_{1,31} & m_{1,32} & m_{1,33} \end{bmatrix} \begin{bmatrix} \tilde{\varepsilon}_{1,t-1} \\ \tilde{\varepsilon}_{2,t-1} \\ \tilde{\varepsilon}_{3,t-1} \end{bmatrix} \\ &+ \begin{bmatrix} m_{2,11} & m_{2,12} & m_{2,13} \\ m_{2,21} & m_{2,22} & m_{2,23} \\ m_{2,31} & m_{2,32} & m_{2,33} \end{bmatrix} \begin{bmatrix} \tilde{\varepsilon}_{1,t-2} \\ \tilde{\varepsilon}_{2,t-2} \\ \tilde{\varepsilon}_{3,t-2} \end{bmatrix} + \dots \end{aligned}$$

If $\tilde{\varepsilon}_{3,t-2}$ increases by 1 unit, then y_{2t} will increase by $m_{2,23}$ units. Hence, the $(2,3)$ -element of $\mathbf{M}_2$, $m_{2,23}$, is the IRF for $y_{2,t+2}$ with respect to $\tilde{\varepsilon}_{3,t}$.

- **The price puzzle:** If we fit a simple recursive VAR using inflation, unemployment, and interest rate using real life data, the IRF will show that a positive shock in r_t will cause an increase in π_t in the short term, contrary to theory.

3.5 Forecasting with VAR

- We still consider a VMA(∞):

$$\mathbf{y}_t = \mathbf{k} + \mathbf{M}_0 \tilde{\mathbf{e}}_t + \mathbf{M}_1 \tilde{\mathbf{e}}_{t-1} + \dots$$

Note that we denote an h -step ahead forecast instead of j for better notation. It follows that:

$$\mathbf{y}_{t+h} = \mathbf{k} + \mathbf{M}_0 \tilde{\mathbf{e}}_{t+h} + \mathbf{M}_1 \tilde{\mathbf{e}}_{t+h-1} + \dots + \mathbf{M}_h \tilde{\mathbf{e}}_t + \mathbf{M}_{h+1} \tilde{\mathbf{e}}_{t-1} + \dots$$

- Therefore:

$$\mathbb{E}[\mathbf{y}_{t+h} \mid I_t] = \mathbf{k} + \mathbf{M}_h \tilde{\mathbf{e}}_t + \mathbf{M}_{h+1} \tilde{\mathbf{e}}_{t-1} + \dots$$

- The forecast error is then:

$$\mathbf{e}_t(h) = \mathbf{y}_{t+h} - \mathbb{E}[\mathbf{y}_{t+h} \mid I_t] = \mathbf{M}_0 \tilde{\mathbf{e}}_{t+h} + \mathbf{M}_1 \tilde{\mathbf{e}}_{t+h-1} + \dots + \mathbf{M}_{h-1} \tilde{\mathbf{e}}_{t+1}$$

- Given that $\Sigma_{\tilde{\epsilon}_t} = \mathbf{I}$, we have the forecast error variance:

$$\Omega_t(h) = \mathbf{M}_0 \Sigma_{\tilde{\epsilon}_{t+h}} \mathbf{M}_0^T + \mathbf{M}_1 \Sigma_{\tilde{\epsilon}_{t+h-1}} \mathbf{M}_1^T + \cdots + \mathbf{M}_{h-1} \Sigma_{\tilde{\epsilon}_{t+1}} \mathbf{M}_{h-1}^T = \mathbf{M}_0 \mathbf{M}_0^T + \mathbf{M}_1 \mathbf{M}_1^T + \cdots + \mathbf{M}_{h-1} \mathbf{M}_{h-1}^T$$

Since the diagonal along the variance-covariance matrix is the variance of each element, we can find the variance for $e_{i,t}(h)$ on the (i, i) -element of Ω_h , which is computed by:

$$[\Omega_t(h)]_{i,i} = \sum_{k=0}^{h-1} \|\mathbf{m}_{k,i}\|^2 = \sum_{k=0}^{h-1} m_{k,i1}^2 + m_{k,i2}^2 + \cdots + m_{k,in}^2$$

where $\mathbf{m}_{k,i}$ is the i -th row vector of $\mathbf{M}_k$.

- In the formula for $\mathbf{e}_t(h)$, we can see that the j -th column of $\mathbf{M}_k$ represents the transmission of the j -th structural shock $\tilde{\epsilon}_j$. By breaking down each column element along the rows, we can conduct a **variance decomposition** to analyze the percentage of how each structural shock explain the variance for each forecast error.
- For example, suppose there are 3 variables, and we want to know how much of the variance of $e_{2,t}(2)$ can be explained by $\tilde{\epsilon}_3$, then we can compute by:

$$\frac{m_{0,23}^2 + m_{1,23}^2}{m_{0,21}^2 + m_{0,22}^2 + m_{0,23}^2 + m_{1,21}^2 + m_{1,22}^2 + m_{1,23}^2}$$

3.6 Model Fitting, Diagnostics, and Evaluation

- Model selection in VAR mainly relies on information criteria, for example:

$$\begin{aligned} \text{MAIC} &= \ln |\hat{\Sigma}| + \frac{2k}{T} \\ \text{MSBIC} &= \ln |\hat{\Sigma}| + \frac{k \ln(T)}{T} \\ \text{MHQIC} &= \ln |\hat{\Sigma}| + \frac{2k \ln(\ln(T))}{T} \end{aligned}$$

where $\hat{\Sigma} = \frac{1}{T} \sum_{t=1}^T \mathbf{e}_t \mathbf{e}_t'$ and k is the number of parameters.

- After fitting a VAR, we conduct diagnostic tests on the residuals, namely:
 1. Test on serially correlated errors: Portmanteau- (e.g. Box-Pierce, Ljung-Box) or Breusch-Godfrey test.
 - H_0 : No serial correlation up to order h .
 2. Test for the structural break: CUSUM test.
 - H_0 : No structural break occurs at time t .
 - Given a time plot, we look at whether there is any time where the cumulative sum of residuals lies outside the boundary of its limiting process.
 3. Test for heteroscedasticity: multivariate ARCH Lagrange-Multiplier test.
 - H_0 : The residuals are homoscedastic.
- After fitting, we are also interested in the relationship between variables.
 - One question to ask: Can the current value of $y_{i,t}$ forecast $y_{j,t+j}$ in the future? Alternatively, Can the past values of $y_{i,t-j}$ forecast $y_{j,t}$?
- Consider a reduced-form VAR(2):

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-2} \\ y_{2,t-2} \end{bmatrix} + \begin{bmatrix} e_{1t} \\ e_{2t} \end{bmatrix}$$

We can conduct a **Granger causality test**, which is an F-test of the hypothesis:

- H_0 : Lagged values of y_{2t} cannot forecast current $y_{1t} \iff y_{2t}$ does not Granger-cause $y_{1t} \iff \phi_{12} = \gamma_{12} = 0$.
- H_1 : Lagged values of y_{2t} can forecast current $y_{1t} \iff y_{2t}$ does Granger-cause y_{1t} .

This is a one-directional test. We may also test on the other way round (H_0 : $\phi_{21} = \gamma_{21} = 0$) or conduct a bi-directional test (H_0 : $\phi_{12} = \gamma_{12} = \phi_{21} = \gamma_{21} = 0$).

- Despite the misleading name, Granger causality does not imply causality, but only the power to forecast.
 - If people buy umbrellas because the sky looks gray in time t , umbrella purchases in time t will Granger-cause rain in time $t + 1$.

4 Non-stationary Time Series

4.1 Deterministic and Stochastic Trend

- We have been talking about stationary processes so far, but in real life, many, if not most, time series are non-stationary in nature (e.g. GDP, stock price).
- First, let z_t denote a stationary process with zero mean ($\mathbb{E}[z_t] = 0$). For example, z_t can be a white noise ($z_t = u_t$) or a stationary AR(1) ($z_t = \phi z_{t-1} + u_t$, $|\phi| < 1$).
- Then, two major categories of non-stationary time series are:
 1. Deterministic trend, e.g. time trend:

$$y_t = \mu + \beta t + z_t$$

Then, $\mathbb{E}[y_t] = \mu + \beta t$ is not constant.

2. Stochastic trend, e.g. unit root process:

$$y_t = \mu + y_{t-1} + z_t$$

As we derived before, $\text{Var}(y_t) = t \text{Var}(z_t)$ which is not constant.

- Each of type has their respective method to detrend into a stationary process:
 1. For a time trend, we can apply **linear detrending** to remove it. Regress y_t against t , and we can obtain the residual:

$$\hat{z}_t = y_t - \hat{\mu} - \hat{\beta}t$$
 which is stationary. Thus, we call it **trend-stationary**.
 2. For a stochastic trend, we conduct **first differencing**:

$$\Delta y_t = y_t - y_{t-1} = z_t$$

which is stationary. Thus, we call it **difference-stationary**.

- What if we apply the wrong method to each other?
 1. Taking first difference of a time trend:

$$\Delta y_t = y_t - y_{t-1} = \beta + z_t - z_{t-1}$$

Although this is stationary, we can see this introduces an artificial MA(1) structure. Then, $\text{Var}(\Delta y_t) > \text{Var}(y_t)$ if $\rho_1 < 0.5$. This is called **over-differencing**.

2. For a unit root process, we expand it as:

$$y_t = \mu + y_{t-1} + z_t = \mu + (\mu + y_{t-2} + z_{t-1}) + z_t = \cdots = \mu t + y_0 + \sum_{j=1}^t z_j$$

Applying linear detrending yields:

$$y_t - \mu t = y_0 + \sum_{j=1}^t z_j$$

Then, we still have $\text{Var}(y_t - \mu t) = t \text{Var}(z_t)$ which is not constant.

4.2 Transitory and Permanent Shocks

- We want to know what is the effect of a short-term shock to the long-term trend.
- For a stationary AR(1): $z_t = c + \phi z_{t-1} + u_t$, we have derived previously that $\mathbb{E}[z_{t+j} | I_t] = c \sum_{k=0}^{j-1} \phi^k + \phi^j z_t$. Then:

$$\frac{\partial \mathbb{E}[z_{t+j} | I_t]}{\partial u_t} = \phi^j \frac{\partial z_t}{\partial u_t} = \phi^j$$

Hence, $\lim_{j \rightarrow \infty} \frac{\partial \mathbb{E}[z_{t+j} | I_t]}{\partial u_t} = 0$, which means a shock has no long-term impact. We call this shock **transitory**.

- Consider a time trend: $y_t = \mu + \beta t + z_t$, where z_t is a stationary AR(1) with zero mean. Note that $\mathbb{E}[z_{t+j} | I_t] = \phi^j z_t$. Then:

$$\mathbb{E}[y_{t+j} | I_t] = \mathbb{E}[\mu + \beta(t+j) + z_{t+j} | I_t] = \mu + \beta(t+j) + \mathbb{E}[z_{t+j} | I_t] = \mu + \beta(t+j) + \phi^j z_t$$

which gives us:

$$\frac{\partial \mathbb{E}[y_{t+j} | I_t]}{\partial u_t} = \phi^j \frac{\partial z_t}{\partial u_t} = \phi^j$$

Therefore, $\lim_{j \rightarrow \infty} \frac{\partial \mathbb{E}[y_{t+j} | I_t]}{\partial u_t} = 0$ for a time trend, so the shock is transitory. This can be attributed to the **trend-reverting** property.

- For a random walk: $y_t = \mu + y_{t-1} + z_t$, where z_t is a stationary AR(1) with zero mean, we can derive:

$$\mathbb{E}[y_{t+1} | I_t] = \mathbb{E}[\mu + y_t + z_{t+1} | I_t] = \mu + y_t + \phi z_t$$

$$\mathbb{E}[y_{t+2} | I_t] = \mathbb{E}[\mu + y_{t+1} + z_{t+2} | I_t] = \mu + (\mu + y_t + \phi z_t) + \phi^2 z_t$$

It follows that:

$$\mathbb{E}[y_{t+j} | I_t] = j\mu + y_t + z_t \sum_{i=1}^j \phi^i$$

Hence,

$$\frac{\partial \mathbb{E}[y_{t+j} | I_t]}{\partial u_t} = \frac{\partial y_t}{\partial u_t} + \sum_{i=1}^j \phi^i \frac{\partial z_t}{\partial u_t} = \frac{\partial z_t}{\partial u_t} + \sum_{i=1}^j \phi^i \frac{\partial z_t}{\partial u_t} = 1 + \sum_{i=1}^j \phi^i = \sum_{i=0}^j \phi^i = \frac{1 - \phi^{j+1}}{1 - \phi}$$

From the decomposition (the first step), we can see that a shock impacts $\mathbb{E}[y_{t+j} | I_t]$ through both y_t and z_t . As $j \rightarrow \infty$, we have:

$$\lim_{j \rightarrow \infty} \frac{\partial \mathbb{E}[y_{t+j} | I_t]}{\partial u_t} = \frac{1}{1 - \phi}$$

which means a one-unit shock to u_t will cause a impact of $\frac{1}{1-\phi}$ units to $\lim_{j \rightarrow \infty} \mathbb{E}[y_{t+j} | I_t]$ in the long run. That is, the shock is **permanent**.

- Historically, economists widely believed that log real GDP and other macroeconomic variables are trend-stationary. However, Nelson and Plosser (1982) could not reject the hypothesis that those follows a unit root process, suggesting that they are difference-stationary. This implied that a shock may have permanent effect on the variables, instead of reverting to a long-term growth path.

- Therefore, we are more interested in difference-stationary trends.

- An example:

- Suppose the long-term trend Y_t^{trend} follows a steady growth rate g , then:

$$Y_t^{\text{trend}} = Y_0^{\text{trend}} \cdot (1 + g)^t \implies \ln Y_t^{\text{trend}} = \ln Y_0^{\text{trend}} + t \cdot \ln(1 + g) \approx \ln Y_0^{\text{trend}} + t \cdot g$$

which is a linear, deterministic trend. We're interested in the true data generating process, where:

$$\ln Y_t = \ln Y_0^{\text{trend}} + t \cdot g + z_t$$

The question is then whether z_t is stationary or a unit root process.

4.3 Difference-stationary and Unit Root Processes

- We denote an d -th difference-stationary process by $I(d)$, which we call it integrated of order d . For example:

- $y_t \sim I(0)$: y_t is a stationary process.
- $y_t \sim I(1)$: y_t is not stationary, but its 1st difference, Δy_t , is stationary.

- Since an $I(d)$ process becomes stationary after taking its d -th difference, the d -th difference can be put into an ARMA, which call it an ARIMA(p, d, q). For example, for an $I(1)$ process, ARIMA($p, 1, q$) is simply an ARMA(p, q) for Δy_t .

- We focus on $I(0)$ and $I(1)$. Furthermore, for $I(1)$, we focus on unit root processes, including the following different cases of random walks:

1. Random walk without drift:

$$y_t = \phi y_{t-1} + z_t$$

2. Random walk with a drift:

$$y_t = \mu + \phi y_{t-1} + z_t$$

3. Random walk with a drift and a time trend:

$$y_t = \mu + \phi y_{t-1} + \beta t + z_t$$

4. Random walk with a drift and a time trend and the trend squared:

$$y_t = \mu + \phi y_{t-1} + \beta_1 t + \beta_2 t^2 + z_t$$

where $\phi = 1$ for all cases if they are random walks.

- To test whether there is a unit root for a dataset, the **Augmented Dickey-Fuller (ADF) test** is suitable for all these cases, where:

- H_0 : The time series has a unit root $\iff \phi = 1$.
- H_1 : $-1 < \phi < 1$
- Note that in an ADF test, the Dickey-Fuller statistic is different from a t -statistic.

- The **KPSS test** puts together trend-stationarity and unit root process, that is:

- H_0 : The time series is stationary or trend-stationary.
- H_1 : The time series has a unit root.

This is helpful in practice for cross-checking with the ADF test result. If the two have contradicting conclusions, then we should further investigate the cause (e.g. structural break, small sample size, near-unit-root cases like $\phi = 0.98$, etc.).

5 Modeling Long-run Relationships: VECM Model

5.1 Cointegration

- $I(0)$ and $I(1)$ processes have the following properties:
 - $I(0) + I(0) = I(0)$
 - $I(0) + I(1) = I(1)$
 - $I(1) + I(1) = I(1)$ **or** $I(0)$, why?
- Let $f_t = f_{t-1} + u_t$, $x_t = f_t + u_{xt}$, $y_t = 2f_t + u_{yt}$, and $z_t = 3f_t + u_{zt}$, then:

$$a_t = x_t + y_t + z_t = 6f_t + u_{xt} + u_{yt} + u_{zt} \sim I(1)$$

However, we can identify some linear combinations, namely:

$$b_t = 2x_t - y_t = 2u_{xt} - u_{yt} \sim I(0)$$

and

$$c_t = 3x_t - z_t = 3u_{xt} - u_{zt} \sim I(0)$$

- Hence, it is possible that some linear combinations of $I(1)$ processes are $I(0)$.
- Formally, suppose $\mathbf{z}_t$ is a vector of $I(1)$ processes, if there exists a **cointegrating relationship** α such that $\alpha^T \mathbf{z}_t \sim I(0)$, then we say the components in $\mathbf{z}_t$ are **cointegrated**.
- As shown in the example above, there can be multiple cointegrating relationships. Specifically, if we have n variables, there can be at most $r = n - 1$ cointegrating relationships, that is, we can have several linearly independent vectors $\beta_1, \beta_2, \dots$ such that $\beta_1^T \mathbf{z}_t, \beta_2^T \mathbf{z}_t, \dots$ are all $I(0)$. Putting the vectors together, we can form a $n \times r$ matrix:

$$\beta = [\beta_1 \quad \beta_2 \quad \dots]$$

Then, $\beta^T \mathbf{z}_t$ yields a vector of r $I(0)$ variables.

- For any linear combinations of $\beta_1, \beta_2, \dots$, namely, any $\alpha \in \text{Col}(\beta)$, we have $\alpha^T \mathbf{z}_t \sim I(0)$, which we can call a **cointegrating space**.
- Empirically, a cointegrating relationship implies a long-term relationship.
 - Murray (1994)'s analogy: Consider a drunk person and their naughty dog. While both of them walk randomly, if they go too far away from each other, they must come back to each other eventually. In this case, we can say their distances, say $y_t - x_t$, where y_t and x_t are their locations, is stationary.
 - In economic theory, one reason of having multiple cointegrating relationships is that we may have different equilibria in different markets.

5.2 VAR in First Differences and VECM

- Suppose $\mathbf{z}_t$ is a vector of $I(1)$ processes. Consider a VAR(3):

$$\mathbf{z}_t = \mu + \Pi_1 \mathbf{z}_{t-1} + \Pi_2 \mathbf{z}_{t-2} + \Pi_3 \mathbf{z}_{t-3} + \mathbf{u}_t$$

To make all components $I(0)$, we can simply take the first difference:

$$\begin{aligned} \Delta \mathbf{z}_t &= \mathbf{z}_t - \mathbf{z}_{t-1} = \mu - \mathbf{z}_{t-1} + \Pi_1 \mathbf{z}_{t-1} + \Pi_2 \mathbf{z}_{t-2} + \Pi_3 \mathbf{z}_{t-3} + \mathbf{u}_t \\ &= \mu + (-\mathbf{I} + \Pi_1) \mathbf{z}_{t-1} + \Pi_2 \mathbf{z}_{t-2} + \Pi_3 \mathbf{z}_{t-3} + \mathbf{u}_t \\ &= \mu + (-\mathbf{I} + \Pi_1 + \Pi_2) \mathbf{z}_{t-1} - \Pi_2 (\mathbf{z}_{t-1} - \mathbf{z}_{t-2}) + \Pi_3 \mathbf{z}_{t-3} + \mathbf{u}_t \\ &= \mu + (-\mathbf{I} + \Pi_1 + \Pi_2 + \Pi_3) \mathbf{z}_{t-1} - (\Pi_2 + \Pi_3) (\mathbf{z}_{t-1} - \mathbf{z}_{t-2}) - \Pi_3 (\mathbf{z}_{t-2} - \mathbf{z}_{t-3}) + \mathbf{u}_t \\ &= \mu + (-\mathbf{I} + \Pi_1 + \Pi_2 + \Pi_3) \mathbf{z}_{t-1} - (\Pi_2 + \Pi_3) \Delta \mathbf{z}_{t-1} - \Pi_3 \Delta \mathbf{z}_{t-2} + \mathbf{u}_t \end{aligned}$$

Denote $\mathbf{\Pi} = -\mathbf{I} + \mathbf{\Pi}_1 + \mathbf{\Pi}_2 + \mathbf{\Pi}_3$, $\mathbf{\Gamma}_1 = -\mathbf{\Pi}_2 - \mathbf{\Pi}_3$, $\mathbf{\Gamma}_2 = -\mathbf{\Pi}_3$, we can write the expression as:

$$\Delta \mathbf{z}_t = \boldsymbol{\mu} + \mathbf{\Pi} \mathbf{z}_{t-1} + \mathbf{\Gamma}_1 \Delta \mathbf{z}_{t-1} + \mathbf{\Gamma}_2 \Delta \mathbf{z}_{t-2} + \mathbf{u}_t$$

This form is called a **vector error correction model (VECM)**.

- In general, for a VAR(p):

$$\mathbf{z}_t = \boldsymbol{\mu} + \mathbf{\Pi}_1 \mathbf{z}_{t-1} + \mathbf{\Pi}_2 \mathbf{z}_{t-2} + \cdots + \mathbf{\Pi}_p \mathbf{z}_{t-p} + \mathbf{u}_t$$

The VECM is:

$$\Delta \mathbf{z}_t = \boldsymbol{\mu} + \mathbf{\Pi} \mathbf{z}_{t-1} + \mathbf{\Gamma}_1 \Delta \mathbf{z}_{t-1} + \cdots + \mathbf{\Gamma}_{p-1} \Delta \mathbf{z}_{t-p+1} + \mathbf{u}_t = \boldsymbol{\mu} + \mathbf{\Pi} \mathbf{z}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{z}_{t-j} + \mathbf{u}_t$$

where $\mathbf{\Gamma}_j = -\sum_{i=j+1}^p \mathbf{\Pi}_i$ and $\mathbf{\Pi} = -\mathbf{I} + \sum_{i=1}^p \mathbf{\Pi}_i$.

- By definition, all components in $\Delta \mathbf{z}_t$ are $I(0)$. Because $I(0) + I(1) = I(1)$ and $u_{i,t} \sim I(0)$, we know that $\mathbf{\Pi} \mathbf{z}_{t-1}$ must be $I(0)$. Then, there are two possibilities:

1. $\sum_{i=1}^p \mathbf{\Pi}_i = \mathbf{I} \implies \mathbf{\Pi} = \mathbf{0}$. Then, the VECM is equivalent to a VAR in first differences:

$$\Delta \mathbf{z}_t = \boldsymbol{\mu} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{z}_{t-j} + \mathbf{u}_t$$

Note that in this case, components in $\mathbf{z}_t$ are not cointegrated.

2. Components in $\mathbf{z}_t$ are cointegrated, where we can decompose by:

$$\mathbf{\Pi} \mathbf{z}_{t-1} = \boldsymbol{\alpha} \boldsymbol{\beta}^T \mathbf{z}_{t-1}$$

where $\boldsymbol{\alpha}, \boldsymbol{\beta}$ are $n \times r$ matrices, and $r = \text{rank}(\mathbf{\Pi})$ is the number of cointegrating relationships.

- If $r = 0$, then it is equivalent to the first case. If $r = n$, then all components are stationary, which means a VAR is sufficient.
 - While $\boldsymbol{\beta}$, the matrix of cointegrating vectors, dictates the long-run relationship, $\boldsymbol{\alpha}$ acts as a matrix of **adjustment coefficients** that controls the direction and magnitude of the variables in the short term.
- For example, consider a simple two-variable case, where:

$$y_t - x_t = \boldsymbol{\beta}^T \mathbf{z}_t = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} \sim I(0) \text{ with zero mean}$$

Let $\boldsymbol{\alpha} = \begin{bmatrix} 0.4 & -0.2 \end{bmatrix}^T$. We consider a simple VECM(0):

$$\begin{bmatrix} \Delta x_t \\ \Delta y_t \end{bmatrix} = \begin{bmatrix} 0.4 \\ -0.2 \end{bmatrix} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} u_{xt} \\ u_{yt} \end{bmatrix} = \begin{bmatrix} 0.4(y_{t-1} - x_{t-1}) + u_{xt} \\ -0.2(y_{t-1} - x_{t-1}) + u_{yt} \end{bmatrix}$$

Suppose there is a positive gap where $y_{t-1} - x_{t-1} = 1$, then $\Delta x_t = 0.4 + u_{xt}$ and $\Delta y_t = -0.2 + u_{yt}$. As x_t is more likely to increase and y_t is more likely to decrease, $y_t - x_t$ is expected to decrease towards the stationary level.

5.3 Test for Cointegration

- How do we test if variables are cointegrated?
- **Engle-Granger test:** If $y_t, x_{1t}, \dots, x_{kt}$ are cointegrated, then in the regression:

$$y_t = \beta_0 + \beta_1 x_{1t} + \dots + \beta_k x_{kt} + u_t$$

u_t should be $I(0)$. In practice, this suggests a two-step approach:

1. Regress y_t against $x_{1t}, \dots, x_{kt}$ and obtain the residual $\hat{u}_t$.
 2. Test whether $\hat{u}_t \sim I(0)$ by ADF test.
- Furthermore, we want to estimate r , the number of cointegrating relations. Then, we can use the **Johansen test**, which has two similar forms of sequential testing:
 1. Trace test:
 - $H_0 : r = 0, H_1 : 0 < r \leq n$
 - $H_0 : r \leq 1, H_1 : 1 < r \leq n$
 - $\vdots$
 - $H_0 : r \leq n-1, H_1 : r = n$
 2. Maximum eigenvalue test:
 - $H_0 : r = 0, H_1 : r = 1$
 - $H_0 : r = 1, H_1 : r = 2$
 - $\vdots$
 - $H_0 : r = n-1, H_1 : r = n$

We test these one by one, and we can get our value of r for the first H_0 not rejected.

5.4 Time Trend in VECM

- It is possible that there is a time trend in a VECM in the deterministic component, i.e. the intercept term, expressed as $\mu_t = \mu_0 + \mu_1 t$, which we can classify five cases:
 1. No constant: $\mu_t = \mathbf{0}$.
 2. Restricted constant: $\mu_t = \mu_0$, where $\mathbb{E}[\Delta \mathbf{z}_t] = \mathbf{0}$. Then, given a VECM:

$$\Delta \mathbf{z}_t = \mu_0 + \alpha \beta^T \mathbf{z}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{z}_{t-j} + \mathbf{u}_t$$

We have:

$$\mathbb{E}[\Delta \mathbf{z}_t] = \mu_0 + \alpha \mathbb{E}[\beta^T \mathbf{z}_{t-1}] = \mathbf{0}$$

Let μ_c be the long-term average $\mathbb{E}[\beta^T \mathbf{z}_{t-1}]$, then:

$$\mu_0 = -\alpha \mu_c$$

3. Unrestricted constant: $\mu_t = \mu_0$, where $\mathbb{E}[\Delta \mathbf{z}_t] \neq \mathbf{0}$.
4. Constant + restricted trend: $\mu_t = \mu_0 + \mu_1 t$, where $\mathbf{z}_t$ has a time trend but $\Delta \mathbf{z}_t$ does not. Then, given a time trend, the VECM looks like:

$$\Delta \mathbf{z}_t = \mu_0 + \mu_1 t + \alpha (\beta^T \mathbf{z}_{t-1} + \rho t) + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{z}_{t-j} + \mathbf{u}_t$$

If $\Delta \mathbf{z}_t$ has no time trend, then the term carrying the time trend t must cancel out each other:

$$\mu_1 = -\alpha \rho$$

5. Constant + unrestricted trend: $\boldsymbol{\mu}_t = \boldsymbol{\mu}_0 + \boldsymbol{\mu}_1 t$, where $\Delta \mathbf{z}_t$ has a time trend.
- Johansen test can test for each of the cases. To determine which to choose, we take the following steps:
 1. Regress $\Delta \mathbf{z}_t$ against t by OLS. If there is a linear trend, then we choose Case 5, otherwise we consider other cases.
 2. Test for $H_0 : \Delta \mathbf{z}_t = 0$. If not rejected, then choose Case 1 or 2, otherwise choose Case 3 or 4.

5.5 Forecasting with VECM

- Consider a VECM(1):

$$\Delta \mathbf{z}_t = \boldsymbol{\mu} + \boldsymbol{\alpha} \boldsymbol{\beta}^T \mathbf{z}_{t-1} + \boldsymbol{\Gamma}_1 \Delta \mathbf{z}_{t-1} + \mathbf{u}_t$$

Then, we have:

$$\begin{aligned} \mathbb{E}[\Delta \mathbf{z}_{t+1} \mid I_t] &= \boldsymbol{\mu} + \boldsymbol{\alpha} \boldsymbol{\beta}^T \mathbf{z}_t + \boldsymbol{\Gamma}_1 \Delta \mathbf{z}_t \\ \mathbb{E}[\Delta \mathbf{z}_{t+2} \mid I_t] &= \boldsymbol{\mu} + \boldsymbol{\alpha} \boldsymbol{\beta}^T \mathbb{E}[\mathbf{z}_{t+1} \mid I_t] + \boldsymbol{\Gamma}_1 \mathbb{E}[\Delta \mathbf{z}_{t+1} \mid I_t] \end{aligned}$$

It follows that:

$$\mathbb{E}[\Delta \mathbf{z}_{t+j} \mid I_t] = \boldsymbol{\mu} + \boldsymbol{\alpha} \boldsymbol{\beta}^T \mathbb{E}[\mathbf{z}_{t+j-1} \mid I_t] + \boldsymbol{\Gamma}_1 \mathbb{E}[\Delta \mathbf{z}_{t+j-1} \mid I_t]$$

Expanding, we have:

$$\mathbb{E}[\mathbf{z}_{t+j} \mid I_t] - \mathbb{E}[\mathbf{z}_{t+j-1} \mid I_t] = \boldsymbol{\mu} + \boldsymbol{\alpha} \boldsymbol{\beta}^T \mathbb{E}[\mathbf{z}_{t+j-1} \mid I_t] + \boldsymbol{\Gamma}_1 \mathbb{E}[\mathbf{z}_{t+j-1} \mid I_t] - \boldsymbol{\Gamma}_1 \mathbb{E}[\mathbf{z}_{t+j-2} \mid I_t]$$

Rearranging, we get a recurrence relation:

$$\mathbb{E}[\mathbf{z}_{t+j} \mid I_t] = \boldsymbol{\mu} + (\mathbf{I} + \boldsymbol{\alpha} \boldsymbol{\beta}^T + \boldsymbol{\Gamma}_1) \mathbb{E}[\mathbf{z}_{t+j-1} \mid I_t] - \boldsymbol{\Gamma}_1 \mathbb{E}[\mathbf{z}_{t+j-2} \mid I_t]$$

where for $j = 1$:

$$\mathbb{E}[\mathbf{z}_{t+1} \mid I_t] = \boldsymbol{\mu} + (\mathbf{I} + \boldsymbol{\alpha} \boldsymbol{\beta}^T + \boldsymbol{\Gamma}_1) \mathbf{z}_t - \boldsymbol{\Gamma}_1 \mathbf{z}_{t-1}$$

Note that this is essentially a VAR(2) in levels. Writing it in the form of $\mathbf{z}_t = \boldsymbol{\mu} + \boldsymbol{\Phi}_1 \mathbf{z}_{t-1} + \boldsymbol{\Phi}_2 \mathbf{z}_{t-2} + \mathbf{e}_t$, we can easily derive the forecast error and variance of forecast error using procedures for VAR.